

SRIRAM DEVATA

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Education

University of Illinois at Urbana-Champaign (UIUC) Aug. 2024 – Present
PhD in Computer Science, Advised by Prof. Sarita Adve 3.83/4

International Institute of Information Technology, Hyderabad (IIITH) Aug. 2019 – July 2024
Bachelor of Technology (Honours) in Computer Science and MS by Research CGPA - 8.99/10

Research/Work Experience

University of Illinois at Urbana-Champaign (UIUC) Aug. 2024 - Present
Graduate Research Assistant

- My broad research focus is to build software and hardware for datacenters.
- Currently working on a serving system for compound AI workloads on datacenter GPUs. Preliminary results show that this serving system decreases the amount of GPU resources needed by 2x.

STARLab, Carnegie Mellon University Jan. 2022 - June 2024
Student Researcher

- Worked with Prof. Akshitha Sriraman and Dr. Gilles Pokam to identify and leverage hardware design opportunities to enhance efficiency in datacenters by optimizing redundant datacenter “tax” operations.
- Developed a methodology that uses perf to profile widely-used open-source microservices to uncover patterns that can guide the development of general hardware accelerators.

Chaincode Labs Sep. 2021 - Jan 2022
Software Developer

- Continuing on work from Summer of Bitcoin, I spearheaded the development of a ‘-nomempool’ runtime option.
- This feature allows for the operation of a Bitcoin node without a mempool, allowing users to participate in the Bitcoin network without needing to mine blocks.

Summer of Bitcoin July 2021 - Sep. 2021
Student Developer

- Streamlined the testing operations of Bitcoin Core by replacing the compiled wallet with a versatile Python mini-wallet.
- Additionally, I also reduced the dependency of a Bitcoin node on its mempool, contributing to the system’s overall robustness and performance optimization.

Relevant Coursework

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| • Manycore Parallel Algorithms (UIUC) | • Introduction to Software Systems (IIIT-H) |
| • Compiler Construction (UIUC) | • Topics in Deep Learning (IIIT-H) |
| • Distributed Systems (UIUC) | • Advanced NLP (IIIT-H) |
| • Advanced Topics in Computer Architecture (UIUC) | • Operating Systems and Networks (IIIT-H) |
| • Parallel Computer Architectures (UIUC) | • Design and Analysis of Software Systems (IIIT-H) |

Technical Skills

Languages: Python, C++, CUDA

Frameworks: PyTorch, RDKit

Utilities/Technologies: Linux, Git, Bash, Slurm, Jupyter, Docker

Awards and Honours

W.J. Poppelbaum Memorial Award (for graduate students in computer architecture), <i>UIUC</i>	2025
Student Travel Grant, <i>MLSys’25</i>	2025
Qualcomm Innovation Fellowship Finalist	2025
CS PhD Fellowship, <i>UIUC</i>	2024
1st Place at <i>ML4Science Hackathon, 2023</i>	2023
Dean’s Merit List for Academic Excellence, <i>IIIT Hyderabad</i>	2019-2021

Grants, Resources, and Funding

Understanding Overheads for Multi-Model Workloads in GPU Datacenters

2025

Received resources with an estimated value of \$85,092 through the NSF ACCESS program.

Selected Projects

GPU Spatial Partitioning for Multi-model Workloads

Jan - May 2025

We evaluated how spatial partitioning on GPUs would impact the working of ML inference pipelines. We found that there is very little overhead due to the additional data movement between spatial partitions, and there is a great advantage in the ML inference latencies being consistent and predictable.

XR Workload Scheduler on a Mobile Device

Jan - May 2025

We developed multiple schedulers that choose to run ML inference on the CPU/GPU/NPU of a mobile XR device to minimize the overall queuing and inference latency. 2 of our schedulers perform better than using only one of the CPU/GPU/NPU in terms of number of deadlines missed.

Distributed File System

Aug - Dec 2024

Implemented a replicated, distributed file system in Golang. The filesystem is resilient and adaptable by implementing features such as heartbeat messages to maintain membership and replication, caching to minimize network traffic, and chunking of large files into smaller blocks.

Desh: a Custom Linux Shell in C

Oct 2021

‘Desh’ is a custom shell designed for Linux-based systems developed from scratch in C, relying exclusively on system calls. This approach required a comprehensive understanding of Linux system architecture and proficient command over C programming, showcasing advanced skills in both system-level programming and operating system concepts.

Professional/Academic Service

Operating Systems Design and Implementation (OSDI’25) Artifact Evaluation Committee

2025

Machine Learning and Systems (MLSys’25) Artifact Evaluation Committee

2025

Compilers, Architecture, and Parallel Computing (CAP) Seminar Organizing Committee, *UIUC*

2025-Present

Other Publications

Generative artificial intelligence for small molecule drug design

August 2024

Current Opinion in Biotechnology

Ganesh Chandan Kanakala*, [Sriram Devata*](#), Prathit Chatterjee, Deva Priyakumar

TorRNA - Improved Prediction of Backbone Torsion Angles of RNA by Leveraging Large Language Models

June 2024

ChemRxiv

[Sriram Devata](#), Deva Priyakumar

DeepSPInN - Deep reinforcement learning for molecular Structure Prediction from Infrared and ¹³C NMR spectra

March 2024

Digital Discovery, Royal Society of Chemistry

[Sriram Devata*](#), Bhuvanesh S.*, Sarvesh M.*, Yashaswi P. , Siddhartha L., Girish V. , Deva Priyakumar